C^{2}DFF-Net for Object Detection in Multimodal Remote Sensing Images

Yue Zhang, Jinbao Chen, Jianyuan Wang, Donghao Shi, Shu Han, and Lixiao Deng

***Abstract*—Multimodal fusion detection has proven to be a practical approach for improving small target perception in remote sensing. Existing studies primarily focus on the fusion of multimodal spatial-domain features, while paying insufficient attention to cross-domain differential features (DFs). Furthermore, exploration of lightweight fusion mechanisms tailored for deployment on edge devices such as UAVs remains relatively limited. To address these issues, we propose an extremely lightweight cross-modal and cross-domain DF fusion network (C^{2}DFF-Net). The network includes three innovative plug-and-play modules. Specifically, we propose a cross-modal DF interaction module (CDFIM) to facilitate intermodal information interaction by enhancing DFs between visible and infrared modalities. To improve the adaptive fusion of multimodal complementary features, we propose a cross-domain gated self-attention (CGSA) module. The global context information of the image is explored from a frequency-domain perspective, and a polarized self-attention (PSA) mechanism is introduced to establish long-range dependencies on spatial-frequency-domain DFs while filtering redundant information. In addition, we design an adaptive light-aware mask (ALM) module to enable the network to learn effective multimodal complementary features without bias, improving its robustness in complex lighting environments such as exposure and glare. Comprehensive experiments demonstrate that C^{2}DFF-Net not only achieves state-of-the-art (SOTA) performance but also maintains a lightweight design, achieving 85.7% mAP50 with 6.58 M parameters and 14.6 giga floating-point operations (GFLOPs) on the DroneVehicle dataset. Furthermore, it shows a favorable accuracy-computation tradeoff compared to SOTA models on the VEDAI and FLIR datasets. Finally, we conduct real-world deployment experiments, and C^{2}DFF-Net achieves satisfactory results under various lighting conditions, clearly demonstrating its practical applicability. Code will be available at https://github.com/FPGAzzy/C2DFF-Net.**

***Index Terms*—Feature fusion, lightweight network, multimodal remote sensing image, spatial-frequency information.**

I. Introduction

With the popularization of high-resolution satellites and uncrewed aerial vehicles (UAVs), object detection in remote sensing images has garnered widespread attention across various fields [1], [2]. Compared to natural image detection, targets in remote sensing images are typically smaller (typically less than 32 × 32 pixels) [3], [4] and face greater challenges in small object detection. These challenges include a low pixel proportion, leading to insufficient feature representation, as well as interference from complex backgrounds and lighting conditions, which cause small targets to blend into the background, thereby reducing recognition accuracy [5].

With the remarkable success of deep learning techniques in the field of natural imagery, numerous studies have successfully extended deep learning algorithms to remote sensing applications [6], [7]. For instance, Zhang et al. [3] introduce the feature enhancement, fusion, and context-aware YOLO (FFCA-YOLO) model, which improves the representation of small targets by incorporating feature enhancement, feature fusion, and spatial context-aware modules. Li et al. [8] propose the cross-stage partial efficient layer aggregation networks (CSP-ELANs) backbone and the converge-and-assign (PCAF) network to effectively address challenges like sparse object pixels and intricate scenes in UAV images. Although visible images offer rich texture and color information, their effectiveness in target monitoring is limited in low-light or nighttime conditions. In contrast, infrared images are unaffected by light but often suffer from low texture, low resolution, and high noise levels. The limitations of single-modality images hinder target detection accuracy and restrict their applicability across all-day, multicondition scenarios. Thanks to the rapid progress in imaging technology, remote sensing images collected from multiple modalities have become feasible. As a result, the integration of visible and infrared images, which provide complementary information, has garnered increasing attention for target detection applications [9], [10], [11]. However, current multimodal fusion techniques still face the following challenges in remote sensing.

A. Inadequate Utilization of Multimodal Complementary Information and Weak Detection of Complex Small Targets in Remote Sensing Images

Specifically, basic fusion methods, such as concatenation or element-wise summation, have been shown in previous studies to be ineffective in leveraging the distinct characteristics of each modality [12]. In addition, due to the unique nature of information from different modalities, cross-modal feature interaction can greatly improve the network's ability to extract features [13]. However, existing feature interaction and

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fusion modules still have room for improvement in learning complementary features across different modalities. In particular, current research has primarily focused on multimodal spatial-domain feature fusion [14], [15], with relatively less attention given to the utilization of cross-domain complementary features, such as the combination of spatial and frequency domains. Consequently, the detection accuracy of complex small targets, which are affected by exposure, glare, density, and lack differentiation from the background, requires further improvement.

B. High Recognition Performance in Multimodal Models Often Comes at the Cost of Significant Computational Effort

Convolutional networks, while effective at capturing local features, struggle with long-range dependencies [16], [17]. To address this issue, researchers have introduced transformer-based fusion techniques, such as Mamba [18], [19] Transformer [20], [21], [22] models, which integrate global contextual information but substantially increase model complexity and computational overhead. These large-scale models are challenging to deploy on small UAV and satellite platforms [15]. As a result, there is still limited exploration of lightweight fusion mechanisms capable of extracting both local and global features for deployment on edge devices.

To overcome the aforementioned challenges, we propose a cross-modal and cross-domain differential feature fusion network (C^2DFF-Net). Our motivation is to propose a lightweight method for small target detection in multimodal remote sensing images, achieving high accuracy through the full interaction and fusion of multimodal features without significant computational overhead. First, we modify the YOLOv8n network into a two-stream backbone as our baseline model. We then propose a cross-modal differential feature interaction module (CDFIM) to enhance intermodality interaction and feature representation. In addition, a cross-domain gated self-attention (CGSA) module is proposed for spatial-frequency information integration. Specifically, to improve the learning of complementary multimodal features, we extract cross-modal and cross-domain differential features (DFs) for interaction and fusion in the two modules. Furthermore, we design an adaptive light-aware mask (ALM) module to apply dynamic complementary masks to multimodal images based on visible local illumination information, accounting for exposure and glare effects, allowing the network to focus effectively on the complementary information from different modalities. Our main contributions are as follows.

1) We propose C^2DFF-Net, a lightweight small target detection method for multimodal remote sensing, while an ALM module is proposed to force the model to learn intermodal complementary features in a balanced way during training.
2) We propose a CDFIM module, which enhances the interaction of complementary information between the two modalities by performing DF extraction and dynamically distributing attention weights across the channel-space dimensions.
3) We propose a CGSA module for feature fusion, which includes a cross-domain feature extraction (CFE) module that efficiently represents complementary information between local spatial and global frequency information. A self-attention feature fusion (SAFF) module enhances complementary features and filters intermodal noise using polarization filtering. An adaptive gating (AG) unit within the SAFF enables deep fusion of multimodal features.
4) Our model is robust for detecting small targets in complex remote sensing images, achieving state-of-the-art (SOTA) performance on the DroneVehicle dataset with a lightweight design [only 6.58 M parameters and 14.6 giga floating-point operations (GFLOPs)]. It also outperforms mainstream models on the VEDAI and FLIR datasets.

The remaining parts of this article are organized as follows. Section II reviews related work in the field of multimodal remote sensing object detection. The proposed C^2DFF-Net architecture is detailed in Section III. Section IV presents ablation studies and comparative experiments with SOTA methods. In addition, real-world deployment experiments are conducted to verify the practical feasibility of C^2DFF-Net. Finally, a summary of our work is provided in Section V.

II. Related Work

A. Multimodal Remote Sensing Object Detection

In the exploration of previous studies, multimodal fusion methods are classified into pixel-level fusion, feature-level fusion, and decision-level fusion, depending on the fusion location.

Pixel-level fusion directly combines multimodal images with low model complexity, such as in SuperYOLO [5] and YOLO-SA [23]. However, due to the lack of interaction between modalities, this method cannot fully leverage multimodal features and is vulnerable to unimodal attacks. Decision-level fusion typically merges the detection results from multiple modalities without explicitly modeling intermodality interactions. Moreover, such methods often involve considerable computational overhead.

In contrast, feature-level fusion, which integrates the fusion steps into the model to form an end-to-end detection framework [14], has garnered more attention from researchers. Qingyun and Zhaokui [12] propose a cross-modality attentive feature fusion (CMAFF) mechanism to extract both shared and unique features from dual modalities. Shen et al. [22] introduce a fusion framework using transformers, called ICAFusion, to model global feature interactions and capture complementary information. Zhao et al. [1] propose a differential multimodal fusion algorithm, called MMFDet, which leverages the complementary characteristics of multimodal data to address the challenging task of detecting small targets in low-light environments. However, the above methods often incur higher computational costs to improve fusion accuracy and capture global contextual relationships. For instance, the ICAFusion model has 120.21 M parameters, and its computational demand reaches up to 191.6 GFLOPs, making it unsuitable for real-time remote sensing detection.

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Surprisingly, some studies have shown that frequency-domain information can capture the global features of images, and its combination with spatial-domain information yields remarkable results. Sun et al. [15] design a lightweight frequency expert encoder (FEE), while Zhou et al. [24] propose a spatial-frequency information integration network (SFINet). Both approaches leverage the Fourier Transform to achieve fast global feature extraction and facilitate the learning of information flow and complementary representations through spatial-frequency-domain interaction.

Therefore, we select the lightweight YOLOv8n model with feature-level fusion as the base detection architecture. In addition, new cross-modal feature interaction and fusion methods are introduced in this article. In particular, inspired by the concept of frequency-domain feature extraction [15], [24], we design the CFE module in CGSA to enhance the characterization of complementary features in multimodal global and local information.

B. Attentional Mechanisms

Attentional mechanisms emphasize the importance of different information by assigning varying weights to feature elements. Integrating these mechanisms into a backbone or feature fusion network can enhance the understanding of fine-grained features and enable the selection of complementary multispectral features [25].

Several typical attention mechanisms have been proposed. SENet [26] utilizes only channel attention, limiting its ability to select important regions. CBAM [27] enhances the model's ability to select key objects and regions by concatenating channel and spatial attention modules. However, sharing information between its spatial structures weakens the adaptation of spatial context to different feature maps. To address this issue, channel-prior convolutional attention (CPCA) [28] introduces a channel-prior attention mechanism and deep strip convolution, which produces dynamically allocated spatial attention maps for each channel, significantly improving segmentation performance.

Acquiring a global receptive field and contextual information is essential to localize small objects. Among attention mechanisms, convolution operations are limited by their local receptive field and can capture long-range dependencies only through deeper convolutions, which increases computational cost. In contrast, the multihead self-attention mechanism in standard transformer frameworks plays a pivotal role in capturing global context in sequence modeling, making it effective for modeling long-range interactions. This ability has made it widely adopted in the vision field, overcoming the limitations of traditional convolutions. However, the multihead self-attention mechanism requires multiple passes through the input data, leading to significant redundancy. To address this, Yang et al. [29] propose replacing multihead self-attention with a multiconvolutional modulator, which reduces redundancy while maintaining global context. Furthermore, polarized self-attention (PSA) [30] uses polarization filtering to compress features in one direction while maintaining high resolution in the orthogonal direction, resulting in high internal resolution in both channel and spatial dimensions. This also reduces computational effort through dimensionality reduction operations. The PSA-GAN model [31] demonstrates that combining the PSA mechanism can reduce prediction error.

Inspired by CPCA and incorporating the cross-modal DF extraction, we propose the CDFIM module to enhance the interaction of complementary features between multispectral modalities. To the best of our knowledge, this is the first application of CPCA from medical image segmentation to multimodal remote sensing detection. Referring to the channel processing method of PSA, we design the SAFF module in CGSA to learn long-term dependencies between multispectral features, eliminate redundant information, and strengthen key information, thereby enabling efficient extraction and thorough fusion of multimodal features.

III. Proposed Method

A. Architecture Overview

The overall architecture of our C²DFF-Net is illustrated in Fig. 1. First, we redesign the feature extraction network of YOLOv8 to form a two-stream backbone that extracts depth features from visible and infrared images separately. In particular, to address the insufficient learning and interaction enhancement of complementary dual-modal information in backbone networks, we design a CDFIM DF interaction enhancement module. This module leverages the differential complementary information between multimodal images to enhance the intermodal features from both the channel and spatial perspectives. In addition, to improve the model's adaptive fusion capability for dual-modal cross-domain DFs and to reduce the computational effort, we propose a CGSA feature fusion module that integrates frequency-spatial-domain information and introduces a lightweight self-attention mechanism to enable remote modeling without increasing complexity. Finally, we retain the original network's detection head structure by fusing multiscale features into the detection head, using the YOLO detector to predict the coordinates, category, and confidence of the target. Aiming to enable the model to learn complementary multimodal features without bias and improve its robustness in complex lighting environments (e.g., exposure and glare), we propose a mask training strategy for ALM based on the illumination prior of the visible image. This strategy balances the model's attention between both modalities while enhancing the learning of effective modal information. Details of each component are shown in Section III.

B. Cross-Modal DF Interaction Module

To enhance the model's ability to learn complementary features from both modalities and improve intermodal information interaction, inspired by the concept of differential amplifiers and CPCA, we propose the CDFIM module as a bridge between the visible and infrared branches. The structure of the CDFIM is illustrated in Fig. 2. Traditional methods for dual-modal information processing, such as direct summation or concatenation, often introduce both differential and redundant information [32]. In contrast, the CDFIM is designed to focus on DFs, thereby enhancing cross-modal complementary features. The CDFIM module consists of two

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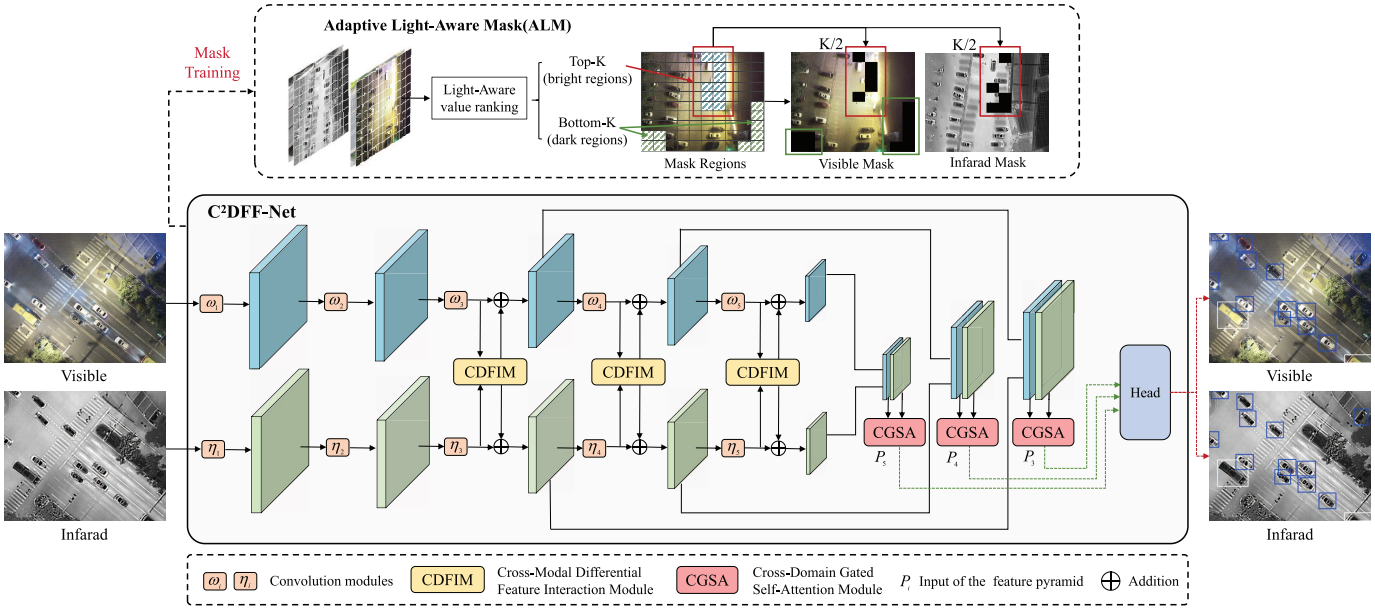


Fig. 1. Framework of the proposed C^{2}DFF-Net. The backbone network consists of three parts: a two-stream feature extraction network, three CDFIM modules to enhance feature interaction, and three CGSA modules for multiscale feature fusion. In addition, we propose the ALM method to train the dual-modal model.

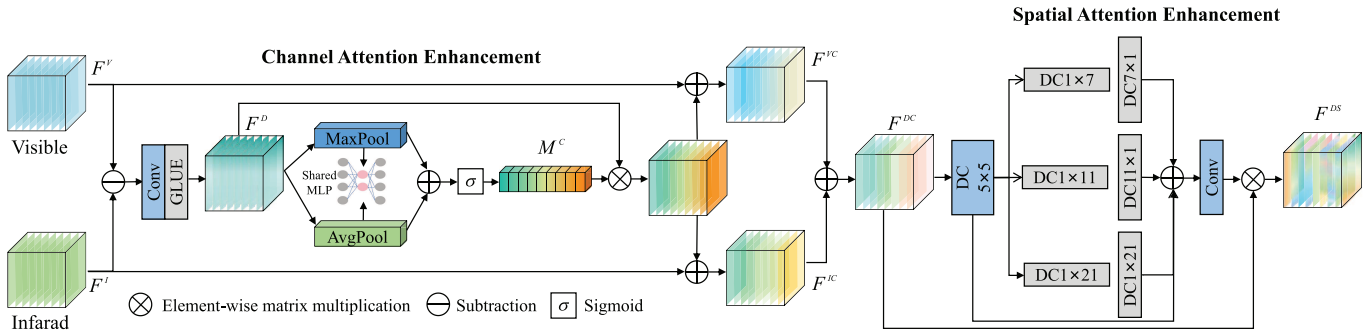


Fig. 2. Overall structure of the CDFIM. Consisting of sequential channel and spatial attention enhancement modules, it enables dynamic attention distribution. The "GMP" module represents global max pooling, while the "GAP" module represents global average pooling. "DC" means depth-wise convolution.

components: channel attention enhancement and spatial attention enhancement. The design dynamically assigns attention weights to both the channel and spatial dimensions, allowing for a more precise focus on the most significant target features.

First, the DF enhancement part. We incorporate the concept of a differential amplifier, where any two-terminal input can be decomposed into a common-mode component and a differential-mode component. The common-mode component represents the average of the two input signals, while the differential-mode component is the difference between them. This circuit amplifies the differential-mode signal while suppressing the common-mode signal, which has been demonstrated in recent multimodal studies. In this part, both the visible convolutional feature map $F^V \in \mathbb{R}^{C\times H\times W}$ and infrared convolutional feature map $F^I \in \mathbb{R}^{C\times H\times W}$ are jointly used as inputs, and the differential-modality feature is obtained by subtracting the two maps. The differential-modality feature is further refined through a 1×1 convolution followed by the GLUE activation function, forming a differential enhancement feature $F^D \in \mathbb{R}^{C\times H\times W}$.

The effect of the differential enhancement module is shown in Fig. 3. It can be observed that the DF maps focus more on the regions overlooked by the single-modal visible and infrared features, indicating that the learning of complementary features between the two modalities has been improved. To facilitate demonstration, we have obtained the additive feature maps by adding the DF maps to the input feature maps of each modality. As shown in the red-marked box, after combining the DFs, both modalities have improved their attention to the vehicle target to varying degrees.

Subsequently, the spatial information within the differential enhancement feature map is aggregated through both average pooling and maximum pooling operations to obtain the global receptive domain. The aggregated spatial information is then processed through a shared multilayer perceptron (MLP), generating the channel attention map $M^C \in \mathbb{R}^{c\times 1\times 1}$. Adaptive

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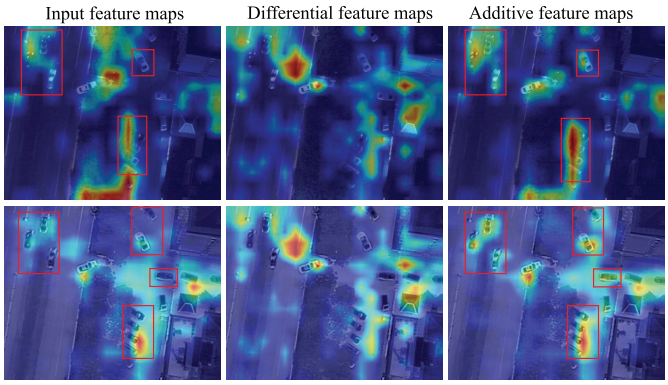


Fig. 3. Illustration of the DF enhancement. The top and bottom rows show attention maps for visible and infrared features, respectively. Input feature maps represent F^V and F^I, DF maps represent F^{D}, and additive feature maps are obtained from $F^V + F^{\mathrm{D}}$ and $F^I + F^{\mathrm{D}}$.

refinement of DFs is achieved by performing element-wise multiplication of F^{D} with M^{C}. Furthermore, inspired by the concept of residual summation, the channel-refined feature map is added to the input feature maps, thereby preserving the integrity of the original features in each modality, denoted as $F^{\mathrm{VC}} \in \mathbb{R}^{\mathrm{C}\times H\times W}$ and $F^{\mathrm{IC}} \in \mathbb{R}^{\mathrm{C}\times H\times W}$. Finally, the two enhanced features are summed to produce the output of the channel attention module, denoted as $F^{\mathrm{DC}} \in \mathbb{R}^{\mathrm{C}\times H\times W}$. This process can be formalized as

$$F^{\mathrm{DC}} = F^{\mathrm{VC}} + F^{\mathrm{IC}} \tag{1}$$

$$F^{\mathrm{VC}} = M^{\mathrm{C}} \otimes F^{\mathrm{D}} + F^{V} \tag{2}$$

$$F^{\mathrm{IC}} = M^{\mathrm{C}} \otimes F^{\mathrm{D}} + F^{I} \tag{3}$$

$$F^{\mathrm{D}} = \mathrm{GLUE}\left(\mathrm{Conv}_1\left(F^{V} - F^{I}\right)\right) \tag{4}$$

$$M^{\mathrm{C}} = \sigma\left(\mathrm{MLP}\left(f_{\mathrm{GAP}}\left(F^{\mathrm{D}}\right)\right) + \mathrm{MLP}\left(f_{\mathrm{GMP}}\left(F^{\mathrm{D}}\right)\right)\right) \tag{5}$$

where $\otimes$ represents element-wise multiplication, σ is the sigmoid function. Conv_1 is 1×1 convolution layer. f_{GAP} and f_{GMP} refer to global average pooling (GAP) and global max pooling (GMP), respectively. MLP stands for the shared MLP. The reduction ratio is set to 4.

After computing the channel attention maps, we employ the depth-separable convolution module to extract the crucial spatial regions for each channel, ultimately generating the dynamically allocated spatial attention maps. Specifically, the spatial attention mechanism follows an Inception-style architecture, which consists of a small kernel convolution to acquire local information, followed by multiple parallel sets of deep strip convolutions to capture contextual information across multiple scales. It is worth noting that a set of strip convolutions can be equivalent to the receptive field of a standard convolution, yet deep strip convolutions are computationally more efficient than 2-D convolutions. In addition, strip convolution facilitates the extraction of features of elongated-shaped objects, such as vehicles, ships, airplanes, etc., which are more suitable for remote sensing targets. The final output feature $F^{DS} \in \mathbb{R}^{\mathrm{C}\times H\times W}$ is obtained by element-wise multiplication of the channel-mixed results and the channel-prior feature, where the channel mixing, using a 1×1 convolution, facilitates the generation of a more refined spatial attention map. The computation of the spatial attention can be described as follows:

$$F^{M} = \mathrm{DwConv}\left(F^{\mathrm{DC}}\right) \tag{6}$$

$$F^{DS} = \mathrm{Conv}_2\left(\sum_{i=0}^{3} \mathrm{Branch}_i\left(F^{M}\right)\right) \otimes F^{\mathrm{DC}} \tag{7}$$

where DwConv denotes depth-wise convolution (DC). Branch_i represents the ith branch of the attention, while the identity connection is indicated by Branch_0. The remaining branches each consist of a set of strip convolutions. Conv_2 is 1×1 convolution layer.

C. CGSA Module

Current multimodal feature fusion methods typically design encoder modules to enhance the reliance on global information or combine convolutional layers with encoders to realize the integrated learning of both local and global information. Inevitably, these approaches come at the cost of computation. To enhance the adaptive fusion of multimodal cross-domain complementary features while reducing the computational cost of the fusion method, we design a lightweight CGSA module for multimodal feature fusion. The CGSA module comprises three main components, as depicted in Fig. 4: First, CFE, which aids in learning complementary features from both the frequency and spatial domains. Second, SAFF facilitates the integration of visible and infrared features, capturing long-range dependencies. Third, an AG unit that dynamically assigns fusion weights, enabling more sensitive and adaptive cross-fusion of features.

1) Cross-Domain Feature Extraction: Leveraging the spectral convolution theorem in Fourier theory, processing data in the Fourier domain enables the extraction of global frequency features, offering a substantial reduction in computational cost compared to traditional encoder-based methods. Let $F^V \in \mathbb{R}^{\mathrm{C}\times H\times W}$ be a visible feature map, and after applying the fast Fourier transform (FFT), the resulting frequency feature map is denoted as $\mathcal{F}(F^V)$, which can be expressed as

$$\mathcal{F}\left(F^{V}\right)(u,v) = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} F^{V}(h,w)\, e^{-2\pi j(uh+vw)} \tag{8}$$

where (h, w) is the position coordinate in the F^V. u and v denote the frequency variables along the h- and W-directions, respectively. In the case of multichannel images, the FFT is applied to each channel individually. For simplicity, we omit the channel notation in the formulas. The amplitude component $\mathcal{A}(F^V)(u, v)$ and phase component $\mathcal{P}(F^V)(u, v)$ are defined as follows:

$$\mathcal{A}\left(F^{V}\right)(u,v) = \sqrt{R^2\left(F^{V}\right)(u,v) + I^2\left(F^{V}\right)(u,v)} \tag{9}$$

$$\mathcal{P}\left(F^{V}\right)(u,v) = \arctan\left(\frac{I\left(F^{V}\right)(u,v)}{R\left(F^{V}\right)(u,v)}\right) \tag{10}$$

where $R(F^V)$ and $I(F^V)$ are the real and imaginary parts of $\mathcal{F}(F^V)$.

Therefore, CFE impacts spatial-domain features by extracting and reconstructing the global frequency information from

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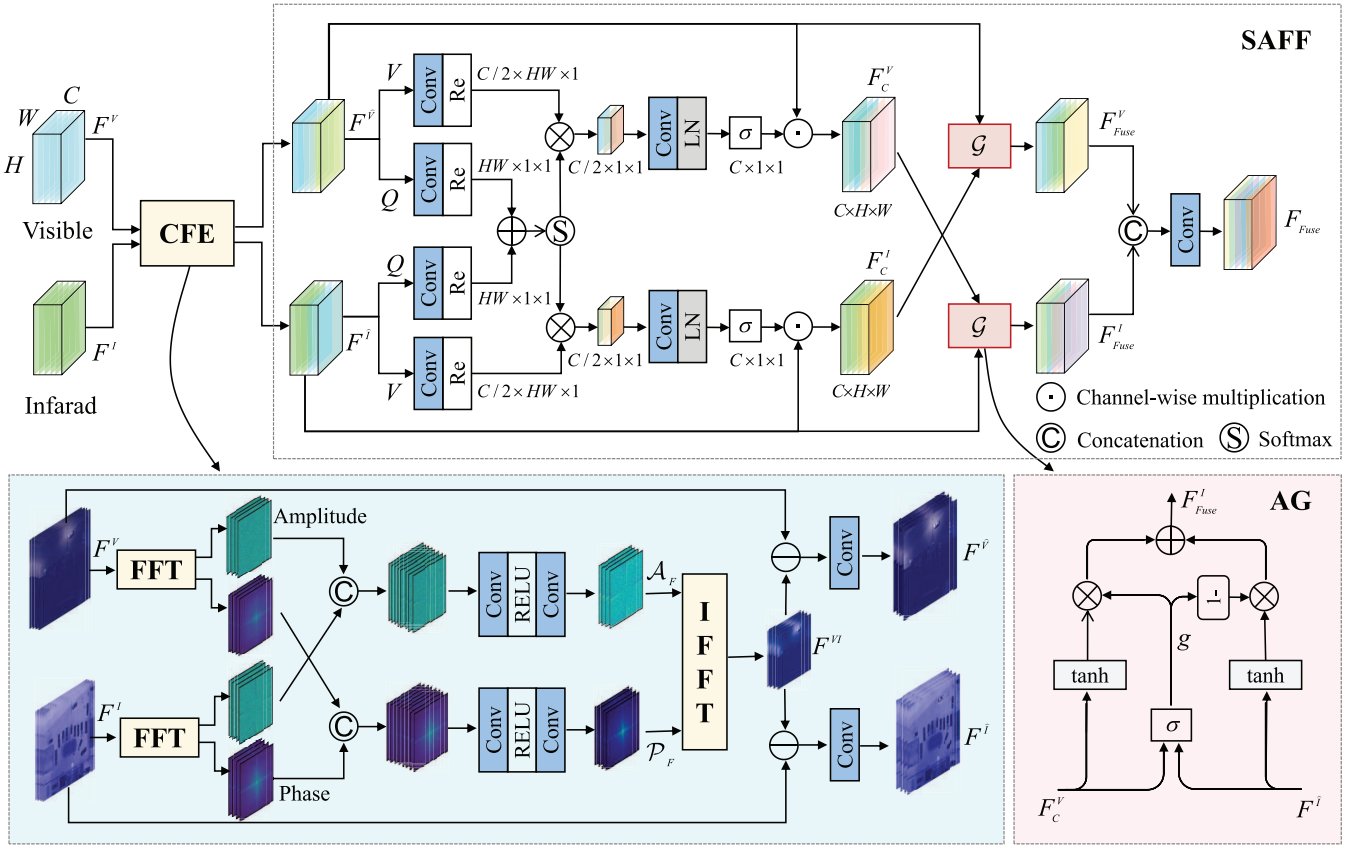


Fig. 4. Detailed architecture of the proposed CGSA module, consisting of three components: CFE, SAFF, and AG unit. "Re" modules represent reshape operations. "LN" modules represent the layer normalization function. The "FFT" module represents the fast Fourier transform, and the "IFFT" module represents the inverse FFT.

both modalities, while differentially enhancing it with spatially localized information. This promotes the flow of information across both the frequency and spatial domains, efficiently filtering and mixing features from different modalities. For the input visible feature map $F^V \in \mathbb{R}^{C\times H\times W}$ and infrared feature map $F^I \in \mathbb{R}^{C\times H\times W}$, we first apply FFT to obtain the amplitude and phase components, denoted as

$$\mathcal{A}\left(F^V\right), \mathcal{P}\left(F^V\right) = \mathcal{F}\left(F^V\right) \tag{11}$$

$$\mathcal{A}\left(F^I\right), \mathcal{P}\left(F^I\right) = \mathcal{F}\left(F^I\right). \tag{12}$$

Then integrate the amplitude and phase information of the two modes separately, followed by two sets of distinct operations to generate an enhanced global frequency representation $\mathcal{A}_F$ and $\mathcal{P}_F$

$$\mathcal{A}_F = f_1\left(\mathrm{Cat}\left[\mathcal{A}\left(F^V\right), \mathcal{A}\left(F^I\right)\right]\right) \tag{13}$$

$$\mathcal{P}_F = f_2\left(\mathrm{Cat}\left[\mathcal{P}\left(F^V\right), \mathcal{P}\left(F^I\right)\right]\right) \tag{14}$$

where Cat[·] represents the channel-wise concatenation operation, f_1 and f_2 serve as a bottleneck layer, consisting of a 1×1 convolution for feature dimension reduction, a ReLU activation function, and another 1×1 convolution for feature dimension expansion.

Next, the inverse FFT (IFFT) is applied to transform the amplitude and phase components resulting from the fusion back into the spatial domain, thereby obtaining the reconstructed feature map $F^{VI} \in \mathbb{R}^{C\times H\times W}$, which is influenced by the frequency domain

$$F^{VI} = \mathcal{F}^{-1}\left(\mathcal{A}_F, \mathcal{P}_F\right) \tag{15}$$

where $\mathcal{F}^{-1}$ IFFT. Finally, we subtract the original spatial-domain features from the features generated in the frequency domain to obtain the cross-domain DFs for each modality as follows:

$$F^{\hat{V}} = \mathrm{Conv}_3\left(F^V - F^{VI}\right) \tag{16}$$

$$F^{\hat{I}} = \mathrm{Conv}_4\left(F^I - F^{VI}\right) \tag{17}$$

where $F^{\hat{V}} \in \mathbb{R}^{C\times H\times W}$ is the cross-domain DF of the visible modality, $F^{\hat{I}} \in \mathbb{R}^{C\times H\times W}$ is the cross-domain DF of the infrared modality. Conv_3 and Conv_4 are 1×1 convolution layers.

2) Self-Attention Feature Fusion: Building on the advantages of self-attention mechanisms in capturing long-range dependencies and internal feature correlations, we propose a novel redesign of the PSA mechanism [30] to guide the fusion of visible and infrared features. Given the spatial alignment of visible and infrared features, our approach emphasizes the fusion of channel-level features across different modalities for optimal integration.

The structure of SAFF is illustrated in Fig. 4. Given the input visible feature $F^{\hat{V}} \in \mathbb{R}^{C\times H\times W}$ and infrared feature

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$F^{\hat{I}} \in \mathbb{R}^{C\times H\times W}$. Take the visible feature as an example. First, $Q^V \in \mathbb{R}^{HW\times 1\times 1}$ and $V^V \in \mathbb{R}^{C/2\times HW\times 1}$ are generated in Q-branch and V-branch consisting of $1*1$ convolution and Reshape operations respectively. In the Q-branch, the feature is fully collapsed, while in the V-branch, the features are retained at high resolution. This can be mathematically expressed as

$$Q^V = \delta_1\left(\text{Conv}_5\left(F^{\hat{V}}\right)\right) \tag{18}$$

$$V^V = \delta_2\left(\text{Conv}_6\left(F^{\hat{V}}\right)\right) \tag{19}$$

where δ_1 and δ_2 represent reshape operations. Conv_5 and Conv_6 denote two distinct 1×1 convolution transformations. Given the symmetry of the structures, the infrared input feature undergoes the same operations, yielding $Q^I \in \mathbb{R}^{HW\times 1\times 1}$ and $V^I \in \mathbb{R}^{C/2\times HW\times 1}$ for the infrared feature.

Next, the Q-branch results of the two modalities are combined with a Softmax operation applied to generate the fused weight distribution. Since the Q-branch features are compressed, the fused weight distribution is multiplied separately by the informative V-branch features of both modalities to enhance the information. Then, pass through a 1×1 convolution layer, followed by layer normalization (LN) and a sigmoid activation function to obtain the attention weights $A^V \in \mathbb{R}^{C\times 1\times 1}$ and $A^I \in \mathbb{R}^{C\times 1\times 1}$. Subsequently, channel-level multiplication is performed between the attention weights and the input feature maps, yielding less noisy feature representations for the visible output feature $F_C^V \in \mathbb{R}^{C\times H\times W}$ and infrared output feature $F_C^I \in \mathbb{R}^{C\times H\times W}$. This process effectively suppresses redundant, non-essential information while emphasizing the critical information for detection. The formula for the visible branch is as follows:

$$A^V = S_g\left[\text{Ln}\left(\text{Conv}_7\left[V^V \otimes S_m\left(Q^V + Q^I\right)\right]\right)\right] \tag{20}$$

$$F_C^V = A^V \odot F^{\hat{V}} \tag{21}$$

where S_m is Softmax function, S_g is Sigmoid function. Conv_7 represents the role of 1×1 convolutional layer. Ln represents LM function. "$\odot$" denotes the channel-wise multiplication operator.

Once the feature representation is filtered, the final feature is obtained by enhancing recalibration through the cross-addition of the two modalities. The formula for the cross-fusion process is as follows:

$$F_{\text{Fuse}}^V = \mathcal{G}\left[F_C^I, F^{\hat{V}}\right] \tag{22}$$

$$F_{\text{Fuse}}^I = \mathcal{G}\left[F_C^V, F^{\hat{I}}\right] \tag{23}$$

where $F_{\text{Fuse}_1}^V$ and $F_{\text{Fuse}_2}^I$ denote the cross-fusion results of the two modalities. $\mathcal{G}$ donates our designed AG.

Finally, after channel-wise concatenation and a 1×1 convolution for channel mixing, the output of the CGSA module is obtained, which represents the fused features of the visible and infrared modalities, denoted as $F_{\text{Fuse}} \in \mathbb{R}^{C\times H\times W}$

$$F_{\text{Fuse}} = \text{Conv}_8\left(\text{Cat}\left[F_{\text{Fuse}}^V, F_{\text{Fuse}}^I\right]\right) \tag{24}$$

where Conv_8 represents the role of 1×1 convolutional layer.

3) AG: Considering that treating dual-modal features equally during fusion may lead to suboptimal performance, we draw inspiration from the gated multimodal unit (GMU) [33] and propose an AG unit for cross-fusion of multimodal features. Specifically, by incorporating a multiplicative gating mechanism and a nonlinear transformation, the proposed AG module adaptively determines the contribution of each modality to the final fusion result based on the characteristics of the input features, effectively filtering out noise. This approach enhances the model's flexibility and dynamic capacity to adjust the importance of different modal features, thereby improving fusion performance and robustness in multimodal tasks. The formula for $\mathcal{G}[F_1, F_2]$ can be expressed as

$$\mathcal{G}[F_1, F_2] = g\cdot\tanh\left(\text{Conv}_9\left(F_1\right)\right) + (1-g)\cdot\tanh\left(\text{Conv}_{10}\left(F_2\right)\right) \tag{25}$$

$$g = S_g\left[\text{Conv}_{11}\left(\text{Cat}\left[F_1, F_2\right]\right)\right] \tag{26}$$

where $\text{Conv}_i, i = \{9, 10, 11\}$ denotes a 1×1 convolution layer. tanh serves as the activation function for a nonlinear transformation. S_g represents the Sigmoid function. g indicates the dynamically learnable weights.

D. Adaptive Light-Aware Mask

When learning dual-modal features using intelligent models, the backbone network typically tends to prioritize the extraction of visible modality features due to the rich texture and color information, inevitably neglecting the infrared modality features [34], [35]. The concept of asymmetric masking can compel the network to learn more effective features from complementary modalities, preventing bias in the model. However, typical random masking strategies do not consider the varying importance of different modalities across different regions of the image. Therefore, to force the model to simultaneously learn dual-modal complementary information and improve its adaptability in complex lighting environments, we propose an ALM module. Unlike the random mask [20] and the DIM [15], our ALM adds dynamic masks for each pair of multimodal images based on the local light-aware information from the visible image. In addition, we consider modality feature learning under glare and exposure conditions, thereby enhancing the model's learning capability in complex environments.

Specifically, as illustrated in Fig. 1, we partition a visible image into $N\times N$ patches, compute the average illumination value for each patch, and subsequently order these patches in ascending sequence based on their illumination values. The bottom-K patches with the lowest rankings are considered as dark regions, while the top-K patches with the highest rankings are considered as bright regions. For the dark regions, the infrared features of targets are generally more pronounced. Therefore, we apply masks to these areas in the visible image, setting the corresponding pixel values to zero, thereby enabling the model to exclusively learn the missing features from the infrared modality. For the bright regions, target contours and color features are more clearly defined in the visible image under normal lighting conditions, making it more suitable for the model to learn from the visible modality in this context. However, the bright regions may also be affected by abnormal

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Fig. 5. Some examples of visible and infrared image pairs from three datasets.

conditions such as glare and exposure. If masks are applied to all corresponding positions in the infrared image, it would be ineffective to solely learn from the visible modality, thereby reducing the model's ability to resist interference. As a result, for the bright regions, we randomly select $K/2$ patch locations within this range to mask the infrared modality, while the remaining $K/2$ locations are masked in the visible modality, to mitigate the impact of glare and exposure conditions on the model. The probability of generating asymmetric masks in the training data is set to 30%.

IV. Experiments

A. Datasets

Our experiments are conducted on the DroneVehicle, VEDAI, and FLIR datasets. Some example images are shown in Fig. 5.

1) DroneVehicle: DroneVehicle [36] is a recently released large multimodal remote sensing vehicle target detection dataset captured by UAVs. The dataset contains 28 439 RGB-infrared image pairs, including urban roads, residential areas, and parking lots, in a variety of complex scenes captured from day to night. Five vehicle types are included, namely "car," "truck," "bus," "van," and "freight car." In our experiments, we have processed the image resolution to 640×512. We use 17 990 image pairs from the official division as the training set, 1469 pairs as the validation set, and 8980 pairs as the test set.

2) VEDAI: VEDAI [37] is a publicly available dataset designed for small target detection in aerial imagery, incorporating challenges such as illumination/shadow variations, glare reflections, and occlusions. The dataset contains 1200 RGB-infrared image pairs with over 3700 annotated targets. Each image contains an average of 5.5 targets, which make up approximately 0.7% of the total pixels. Nine types of ground targets are included: car, pickup, camping, truck, other, tractor, boat, van, and plane. In this article, we use the 1024×1024 pixel version for experiments, selecting 1089 image pairs for the training set and 121 image pairs each for the validation and test sets.

3) FLIR: The FLIR dataset [38] is a widely used benchmark dataset for evaluating multispectral target detection, including a large number of RGB and thermal images, including day and night scenes. In our experiments, we use the aligned FLIR dataset [39], which is aligned according to the original data. There are 5142 image pairs in this dataset, of which 4129 pairs are used for training and 1013 pairs are used for testing. This dataset covers three object categories: namely "person," "car," and "bicycle."

B. Implementation Details

1) Baseline: We construct a dual-channel detection network as the base detector by incorporating an infrared feature extraction branch into YOLOv8n, utilizing concatenation as the feature fusion method.

2) Experiment Settings: All our experiments are conducted on the Windows 11 operating system, with the programming environment set up in PyCharm IDE using Python 3.8, PyTorch 1.8, and CUDA 11.1. The computing resources include an Intel Core i7-13700KF CPU and an NVIDIA RTX 4090 GPU. The training phase takes 300 epochs with a batch size of 16. The SGD optimizer is used with the initial learning rate of 0.01, the momentum of 0.937, and the weight decay is 0.0005. For the DroneVehicle and FLIR datasets, both training and testing utilize images resized to 640×640. For the VEDAI dataset, the resolution is set to 1024×1024.

3) Evaluation Metrics: Model accuracy is evaluated using PASCAL-style average precision (AP) with IoU thresholds from 0.5 to 0.95, and AP50 at an IoU threshold of 0.5. In addition, COCO-style APs are reported for small objects (less than 32×32). For model efficiency, GFLOPs and parameter size are used.

C. Ablation Study

1) Validation of the Baseline Framework: The baseline used in the experiments is a dual-stream YOLOv8n detector that adopts middle-level fusion with channel concatenation. To demonstrate the rationality of our baseline model, we compare the performance of lightweight versions of YOLOv5 [40], YOLOv8 [41], and YOLOv11 [42] under various input modalities, fusion levels, and fusion methods using the DroneVehicle dataset. As shown in Table I, dual-modal fusion outperforms single modality for the task of remote sensing small target

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TABLE I
COMPARISON RESULTS OF DIFFERENT MODELS IN THE YOLO FRAMEWORK ON THE DRONEVEHICLE DATASET

Method	Input Type	mAP50	Params	GFLOPs
YOLOv5n	RGB	71.9	1.78M	4.3
YOLOv5n	IR	74.0	1.78M	4.3
YOLOv8n	RGB	72.2	3.01M	8.2
YOLOv8n	IR	79.6	3.01M	8.2
YOLOv11n	RGB	70.0	2.60M	6.5
YOLOv11n	IR	80.8	2.60M	6.5
YOLOv5n+ MidFusionCAT	Multi	74.3	3.37M	8.9
YOLOv8n+ MidFusionCAT	Multi	**82.9**	4.40M	11.5
YOLOv11n+ MidFusionCAT	Multi	81.9	3.29M	8.1

TABLE II
COMPARISON RESULTS OF YOLOV8N WITH PIXEL-LEVEL AND FEATURE-LEVEL FUSIONS ON THE DRONEVEHICLE DATASET

Method	Input Type	mAP50	Params	GFLOPs
InputFusionAdd	Multi	79.8	3.01M	8.1
InputFusionCAT	Multi	82.0	3.01M	8.2
InputFusionMF	Multi	81.7	3.01M	8.8
MidFusionAdd	Multi	82.0	4.28M	11.2
MidFusionCAT	Multi	82.9	4.40M	11.5
MidFusionMF	Multi	**83.1**	5.99M	14.4

TABLE III
EFFECTS OF CDFIM MODULE ON THE DRONEVEHICLE DATASET

Method	mAP50	Params	GFLOPs
Baseline	82.9	4.40M	11.5
CPCA	83.8	5.18M	13.7
CPCA+DF	84.3	4.66M	12.2
CPCA+DF+RA(CDFIM)	**84.7**	4.66M	12.2

detection. Among the dual-modal models, our baseline model achieves 8.6% and 1% higher mAP50 compared to YOLOv5 and YOLOv11, respectively.

Table II presents a comparison of pixel-level (denoted as InputFusion) and feature-level (denoted as MidFusion) fusion in the dual-modal setting, with fusion methods including summation, channel concatenation, and the MF module in SUPERYOLO [5]. It can be observed that feature-level generally yields higher recognition accuracy than pixel-level fusion under the same fusion method. This indicates that feature fusion within the model is more effective at fully learning bimodal feature information than image fusion at the input stage. For our baseline, although the mAP is slightly lower than that of the MidFusionMF method, the number of parameters and computational complexity are reduced by 36.1% and 25.2%, respectively. Consequently, this makes it more practical to design a lightweight multimodal model. These findings support the validity of our baseline detection framework.

2) Impact of CDFIM Module: We conduct an ablation study of the CDFIM to evaluate the impact of this feature interaction module on the model's performance. The CDFIM is built upon the CPCA mechanism [28] and incorporates two key components: a DFDF enhancement module and a residual addition (RA) strategy. As shown in Table III, the experimental results indicate that mAP improves by 0.9% after the introduction of the CPCA attention mechanism compared to the baseline model. The CPCA method in the experiment is implemented by directly concatenating the dual-modal feature channels. The addition of the DF module further enhances mAP by 1.4% while simultaneously reducing computational complexity and parameter count. The RA module contributes to more stable feature learning through its preservation of original feature information. Ultimately, the addition of the CDFIM to the baseline model leads to a 1.8% improvement in mAP, while the parameter count increases by only 0.26 M and the computation cost rises by just 0.7 GFLOPs. This demonstrates the effectiveness of the CDFIM.

TABLE IV
EFFECTS OF CGSA MODULE ON THE DRONEVEHICLE DATASET

Method	mAP50	Params	GFLOPs
Baseline	82.9	4.40M	11.5
PSA	84.3	4.67M	12.1
SAFF	84.4	4.67M	11.8
SAFF+AG	84.7	5.01M	12.4
SAFF+CFE	84.6	6.22M	13.3
SAFF+CFE+AG(CGSA)	**84.9**	6.31M	13.9

3) Impact of CGSA Module: We conduct ablation experiments on the CGSA module, as shown in Table IV. By replacing the concatenation fusion method with the SAFF module, we observe a 1.5% improvement in mAP50 with only minimal computational overhead. Since SAFF is a modification of the channel module in the PSA attention mechanism [30], we compare the performance of SAFF against the full PSA attention mechanism to demonstrate the effectiveness of using only the channel module. We have modified the PSA mechanism to a dual-modal input in our experiments. It is observed that without using the spatial attention mechanism, recognition accuracy improves, and computational effort is reduced. This improvement may be due to the fact that there is little spatial misalignment between modes in the multimodal dataset. Furthermore, when we integrate our CFE and AG modules into SAFF, recognition accuracy improves by 1.7% and 1.8%, respectively. This demonstrates the effectiveness of incorporating frequency-domain global features for extracting cross-domain DFs and AG units for dynamically learning feature fusion weights. Finally, the introduction of the CGSA module results in a 2.0% performance improvement over the original model, confirming its ability to enhance the deep fusion of features from different modalities.

To quantitatively demonstrate the redundancy-reducing effect of the CGSA module, we compare the interchannel Pearson correlation coefficient matrices of the feature maps from the same layer across different models, as shown in Fig. 6. These feature maps are obtained from the baseline model and its variants, which incorporate the SAFF and CGSA

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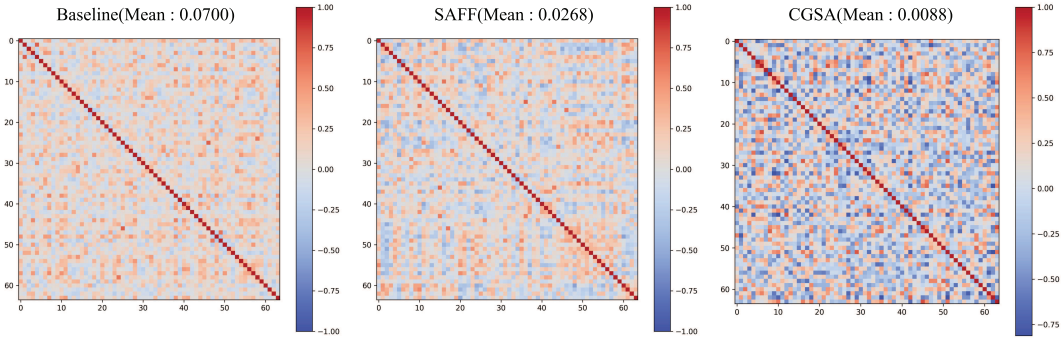


Fig. 6. Pearson correlation coefficient matrix of the interchannel feature maps for layer 32 of the model. "Mean" represents the average correlation, calculated as the mean of the absolute values of the off-diagonal elements of the correlation coefficient matrix.

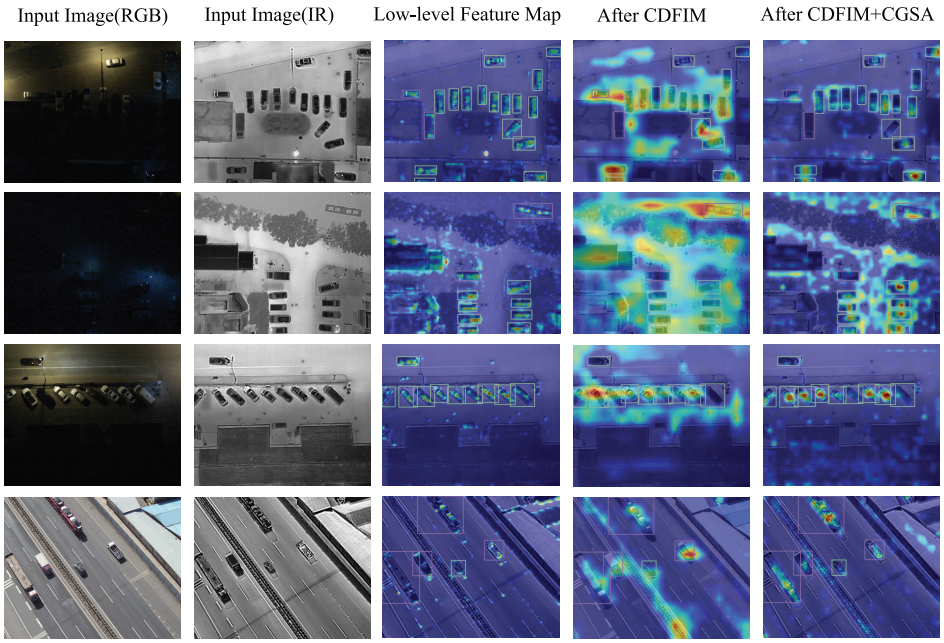


Fig. 7. Influence of CDFIM and CGSA on feature extraction. Brighter colors highlight areas of greater model attention.

modules at layer 32. This layer marks the first point in the model where dual-modal feature fusion is performed. It can be observed that, compared to the channel concatenation fusion method, the average correlation values between the CGSA module and its core SAFF module are lower. This suggests that the information independence between channels in the feature map is enhanced, effectively suppressing redundant feature information. This improvement is attributed to the introduction of the polarization filtering mechanism in the SAFF module, along with the CFE module in CGSA, which extracts frequency-domain DFs.

4) Impact of the Attentional Mechanisms: To assess the effectiveness of our designed core modules in enhancing detection performance, we present the feature maps of the model at various stages: at the low-level, after applying CDFIM, and after incorporating CDFIM+CGSA in Fig. 7. The feature maps in the same columns correspond to identical layer positions. The results demonstrate that, with the introduction of the CDFIM and CGSA modules, the model shows a stronger focus on the target while effectively suppressing background noise. These observations indicate that our designed modules significantly improve the model's ability to concentrate on the target.

TABLE V
INFLUENCE OF THE PATCH NUMBER N AND TOP/BOTTOM-K OF ALM ON THE DRONEVEHICLE DATASET

Number	Probability	mAP50	mAP
/	/	85.2	64.6
N4K4	0.3	84.8	64.6
N6K4	0.3	84.9	64.4
N8K4	0.3	**85.7**	**65.2**
N8K6	0.3	84.4	63.9
N8K8	0.3	85.4	64.6
N10K6	0.3	85.5	65.2
N10K8	0.3	85.2	64.5
N10K10	0.3	84.9	64.8

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TABLE VI
RESULTS OF THE ABLATION STUDY FOR DIFFERENT ESSENTIAL COMPONENTS IN OUR MODEL ON THREE DATASETS

Dataset	CDFIM	CGSA	ALM	mAP50	mAP	APs	Param	GFLOPs
DRONE				82.9	62.3	23.0	4.40M	11.5
	✓			84.7	64.3	28.8	4.66M	12.2
		✓		84.9	64.4	25.6	6.31M	13.9
	✓	✓		85.2	64.6	32.1	6.58M	14.6
	✓	✓	✓	**85.7**	**65.2**	**32.3**	6.58M	14.6
VEDAI				72.0	44.1	38.1	4.40M	11.5
	✓			73.5	46.0	39.6	4.66M	12.2
		✓		78.6	49.6	42.1	6.31M	13.9
	✓	✓		79.4	50.0	43.0	6.58M	14.6
	✓	✓	✓	**79.8**	**50.2**	**43.8**	6.58M	14.6
FLIR				73.3	38.8	27.7	4.40M	11.5
	✓			75.2	39.4	28.0	4.66M	12.2
		✓		75.0	40.0	28.4	6.31M	13.9
	✓	✓		75.8	39.9	28.6	6.58M	14.6
	✓	✓	✓	**76.9**	**40.8**	**29.2**	6.58M	14.6

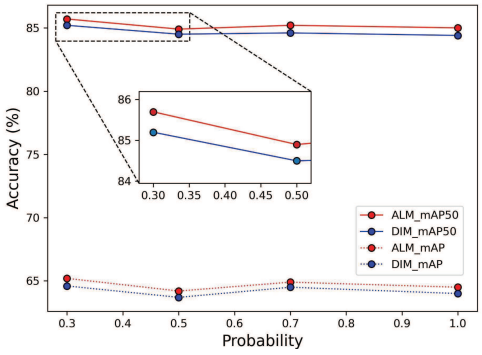


Fig. 8. Comparison of DIM and ALM in various process probabilities.

5) Impact of ALM: For the ALM, Table V explores the effect of different numbers of division patches (N) and top/bottom-K on the accuracy of our C²DFF-Net model, with Probability denoting the percentage of mask processing in the training set, uniformly set to 0.3. It can be seen that when N is set to 8 and K is set to 4, the model's mAP50 improves by 0.5%, reaching a maximum value of 85.7%. This demonstrates that the ALM method is an effective training strategy. In addition, a larger mask area does not necessarily lead to better performance. This is because ALM essentially introduces noisy data to force the network to learn complementary modal information. When there is too much noisy data, it negatively impacts the network's performance.

Furthermore, we compare ALM with another light-aware masking method called DIM [15]. As shown in Fig. 8, our ALM consistently performs better for the same combination of parameters and processing probabilities. This is because, compared to DIM, we account for the possibility of exposure and glare in bright regions. Masks are added to the visible mode at half of the random positions in bottom-K, while masks are added to the remaining positions in the infrared mode. This strategy avoids relying solely on visible modal information during poor lighting conditions, which could not only provide invalid information but also introduce more noise. Moreover, different data processing probabilities affect the model's recognition accuracy. The best performance is achieved when the processing probability is 0.3. Therefore, in subsequent experiments, we set the ALM processing probability to 0.3 across different datasets.

6) Overall Impact of Each Module: To reveal the contributions of each block in the detector, we perform ablation experiments on three datasets: DroneVehicle, VEDAI, and FLIR. Note that the baseline is based on the devised YOLOv8n+MidFusionCAT for fair comparisons. We apply CDFIM, CGSA, and ALM step by step to the baseline to verify their validity. Table VI shows the effect of adding each module on the evaluation metric, where ✓ indicates the use of the module.

It can be found that on the three datasets, DroneVehicle, VEDAI, and FLIR, the mAP50 of the model improves by 1.8%, 1.5%, and 1.9% when the CDFIM module is added, and the mAP50 improves by 2.0%, 6.6%, and 1.7% when the CGSA module is included. This indicates that the complementary learning ability of dual-modal features and feature fusion can be effectively enhanced by our CDFIM and CGSA modules. When both modules are deployed simultaneously, the model achieves even better performance, with mAP50 increasing by 2.3%, 7.4%, and 2.5%, and APs improving by 9.1%, 4.9%, and 0.9%, respectively. The limited improvement in small target recognition accuracy on the FLIR dataset is due to the fact that this dataset primarily focuses on target detection in complex urban environments, where small target samples constitute a relatively small proportion. It is worth noting that our parameter count increases by only 2.18 M, and computational cost increases by only 3.1 GFLOPs. This improvement is attributed to the cross-modal DFs and cross-domain DFs extracted in our two modules, which avoid learning redundant information. In addition, we introduce deep strip convolution, FFT, and a PSA mechanism to achieve feature extraction enhancement with very low computational effort. Finally, when combined with the ALM training strategy,

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TABLE VII
PERFORMANCE COMPARISON OF DIFFERENT DETECTION METHODS ON THE DRONEVEHICLE DATASET

Method	Modality	car	freight	truck	bus	van	mAP50	mAP	Param	GFLOPs
YOLOv8n [41]	RGB	90.8	54.8	67.3	91.6	56.5	72.2	45.4	3.01M	8.1
PCAF-N [8]	RGB	95.0	50.6	65.4	94.5	58.7	72.8	50.2	10.20M	46.2
PCAF-L [8]	RGB	98.2	67.2	78.9	96.7	67.4	81.7	57.3	57.90M	483.2
FFCA-YOLO [3]	RGB	92.1	56.5	69.6	93.7	60.8	74.5	45.6	7.12M	51.2
YOLOv8n [41]	IR	98.1	66.4	74.0	96.8	62.5	79.6	59.1	3.01M	8.1
PCAF-N [8]	IR	98.2	67.8	70.9	96.0	58.9	78.4	56.6	10.20M	46.2
PCAF-L [8]	IR	98.3	77.0	80.2	**97.5**	69.5	**84.5**	61.5	57.90M	483.2
DTNet-B [43]	IR	90.2	67.9	78.1	89.2	65.7	78.2	52.9	27.67M	57.1
ERGW-net [44]	IR	96.9	74.8	77.9	96.1	66.8	82.5	-	-	-
FFCA-YOLO [3]	IR	98.2	68.8	77.9	96.9	66.1	81.6	60.5	7.12M	51.2
LF-MDet [15]	Multi	82.2	59.6	73.6	86.6	57.0	71.8	51.3	38.70M	77.7
E2E-MFD [11]	Multi	90.3	64.4	79.3	89.8	63.1	77.4	-	-	-
C^2Former [10]	Multi	90.2	64.4	68.3	89.8	58.5	74.2	-	-	100.9
ICAFusion [22]	Multi	96.8	61.1	74.4	95.3	49.5	75.4	52.6	120.21M	191.6
SuperYOLO [5]	Multi	**98.4**	69.3	**95.9**	65.5	64.4	78.7	56.1	4.85M	56.0
OAFA [9]	Multi	90.3	73.3	76.8	90.3	66.0	79.4	-	-	-
MGMF [45]	Multi	91.4	**78.5**	70.1	91.1	69.4	80.3	55.2	122.00M	-
MMFDet [1]	Multi	80.7	**98.4**	**97.2**	66.7	69.3	82.5	-	3.50M	>150.6
YOLO-SA [23]	Multi	98.2	76.4	78.0	96.8	66.4	84.2	-	-	-
Baseline	Multi	**98.4**	72.4	79.3	97.0	**72.4**	82.9	**62.3**	4.40M	11.7
Ours	Multi	**98.6**	77.3	84.1	**97.6**	**70.8**	**85.7**	**65.2**	6.58M	14.6

TABLE VIII
PERFORMANCE COMPARISON OF DIFFERENT DETECTION METHODS ON THE VEDAI DATASET

Method	Modality	car	pickup	camp	truck	other	tractor	boat	van	plane	mAP50	mAP	Param	GFLOPs
YOLOv8n [41]	RGB	86.9	82.7	67.3	73.0	61.8	73.1	58.8	77.5	58.0	71.0	45.4	3.01M	8.1
FFCA-YOLO [3]	RGB	89.3	**86.7**	76.9	74.7	51.0	78.5	49.7	74.6	63.8	71.7	44.8	7.12M	51.2
SuperDet [46]	RGB	88.6	82.4	77.3	70.0	47.2	77.9	64.1	**93.8**	**99.8**	77.6	-	-	-
YOLOv8n [41]	IR	85.4	78.8	72.8	**81.9**	31.1	49.1	57.0	87.2	99.5	71.4	46.0	3.01M	8.1
FFCA-YOLO [3]	IR	85.5	86.3	81.4	74.8	29.8	68.1	59.9	**96.2**	35.9	68.6	42.1	7.12M	51.2
CM-YOLO [47]	IR	86.7	82.5	**83.7**	74.4	50.6	68.5	54.8	78.9	-	72.5	-	16.97M	24.92
DMM [18]	Multi	84.2	78.8	79.0	64.7	56.2	72.3	**72.3**	72.5	93.6	75.0	-	87.97M	-
MMFDet [1]	Multi	88.3	78.5	81.6	59.8	63.5	**86.2**	**76.0**	88.3	-	77.9	-	3.50M	>150.6
YOLOFusion [12]	Multi	**91.7**	85.9	78.9	78.1	54.7	71.9	71.1	75.2	**99.5**	78.6	49.1	24.50M	34.90
ICAFusion [22]	Multi	84.4	82.5	73.4	76.2	**69.5**	59.9	53.0	81.9	-	72.6	41.6	120.21M	191.6
SuperYOLO [5]	Multi	90.0	86.8	82.2	80.8	60.7	67.0	72.1	72.5	76.5	76.5	47.6	4.85M	56.0
MMYFnet [48]	Multi	90.8	**88.3**	74.5	**84.9**	**72.6**	74.9	60.0	74.6	99.5	**80.0**	**52.1**	17.16M	43.0
Baseline	Multi	89.2	83.6	78.0	74.8	61.8	71.9	57.0	71.3	60.6	72.0	44.1	4.40M	11.5
Ours	Multi	**91.8**	85.9	**87.2**	74.4	58.7	**78.6**	62.2	91.3	77.5	**79.8**	**50.2**	6.58M	14.6

the recognition accuracy improves to 85.7%, 79.8%, and 76.9% on the three datasets, respectively, without adding any computational burden to the model. For the parameter combinations of ALM, we set them to N8K4, N10K6, and N10K6 on the three datasets, respectively.

The above results demonstrate that each of our modules makes a positive contribution to overall performance and is well-suited for small target detection, as well as for different scenarios such as remote sensing and autonomous driving.

D. Comparison With SOTA Methods

1) Quantitative Evaluation: We compare the proposed method with several SOTA methods on the comparison results of the DroneVehicle, VEDAI, and FLIR datasets. The results demonstrate the significant advantages of our algorithm in terms of accuracy and lightweight.

Table VII presents our experimental results on the DroneVehicle dataset. Our methods successfully surpass other SOTA methods on an extremely lightweight scale, achieving a recognition accuracy of 85.7% in mAP50 with only 6.58 M parameters and 14.6 GFLOPs of computation. The infrared model of PCAF-L [8] has the second-best detection accuracy and even outperforms some multimodal fusion methods. However, its parameter count and computational cost are 8.8× and 33.1× greater than ours, and it is limited to single-modality detection. Among the prior multimodal detection models, MMFDet [1] shows the closest recognition accuracy, which is 3.2% lower than ours. However, it does not achieve the simultaneous reduction in parameters and computational load, with its computational cost being over 10× higher than ours.

Table VIII presents our experimental results on the VEDAI dataset. Our C^2DFF-Net outperforms the multimodal baseline by 7.8% in mAP50, demonstrating the effectiveness and efficiency of the proposed module. Our model achieves the highest single-class detection accuracy for targets in the car

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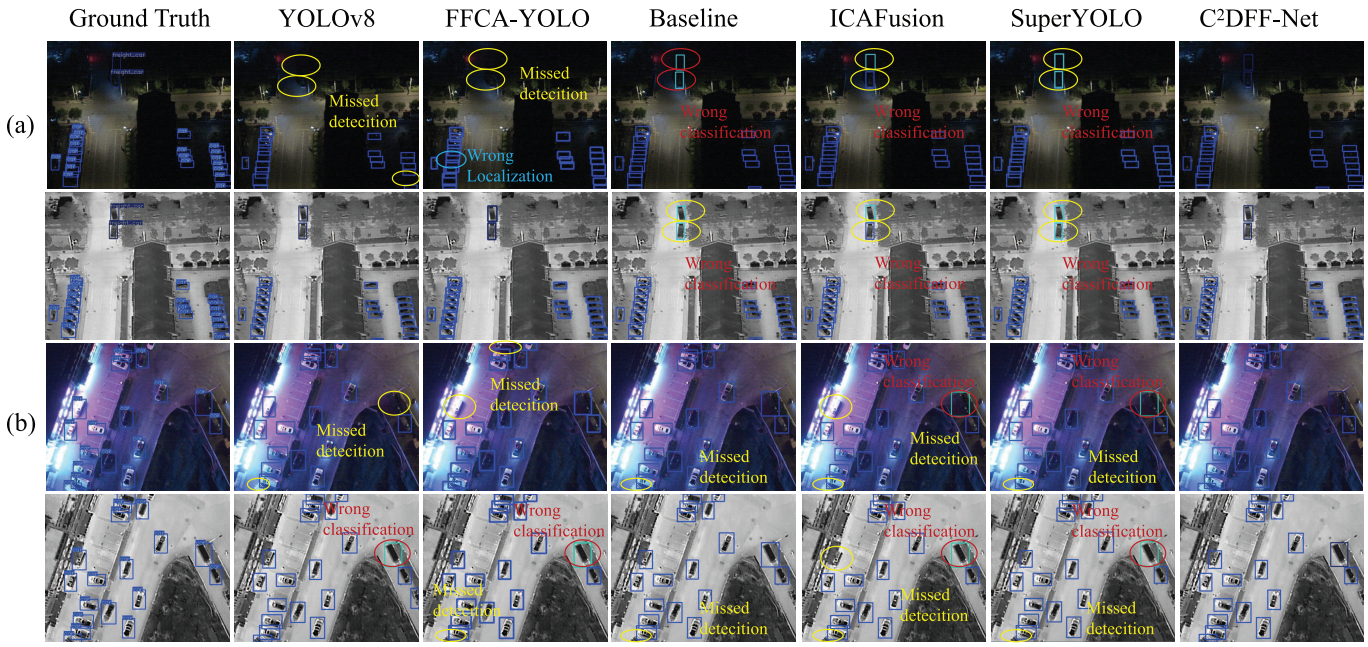


Fig. 9. Visualized detection results of different models on the DroneVehicle dataset. For YOLOv8 and FFCA-YOLO, the first and second rows correspond to visible and infrared single-modality detection results, respectively. The remaining rows present the results of bimodal detection models. (a) Some targets are completely submerged in darkness at night. (b) Overexposure occurs in several regions.

TABLE IX
PERFORMANCE COMPARISON OF DIFFERENT DETECTION METHODS ON THE FLIR DATASET

Method	Modality	mAP50	mAP	Param	GFLOPs
YOLOv8n [41]	RGB	61.6	28.9	3.01M	8.1
YOLOv11n [42]	RGB	60.1	28.1	2.58M	6.3
FFCA-YOLO [3]	RGB	62.4	28.6	7.12M	51.2
YOLOv8n [41]	IR	73.0	37.6	3.01M	8.1
YOLOv11n [42]	IR	72.5	37.3	2.58M	6.3
FFCA-YOLO [3]	IR	74.8	38.7	7.12M	51.2
SuperYOLO [5]	Multi	69.3	33.8	4.84M	55.9
YOLOFusion [12]	Multi	76.6	39.1	24.50M	34.90
MMYFnet [48]	Multi	76.8	**40.9**	17.16M	43.0
ICAFusion [22]	Multi	**77.0**	33.7	120.21M	191.6
GAFF [49]	Multi	72.9	37.5	23.77M	-
YOLO-MS [14]	Multi	75.2	38.3	15.30M	36.9
Baseline	Multi	73.3	38.8	4.40M	11.5
Ours	Multi	**76.9**	**40.8**	6.58M	14.6

category, highlighting its superior performance in detecting small targets in remote sensing. Compared to the SOTA model MMYFnet [48], although our model has a slight decrease in mAP by 0.2%, it reduces the parameter count by 61.7% and computational cost by 66.0%. Notably, the lightweight nature of our model is advantageous for practical deployment and fast inference.

Table IX presents our experimental results on the FLIR dataset. Unlike remote sensing datasets, FLIR is a more generalized, large-scale urban dataset. Our method achieves 76.9% and 40.8% in mAP50 and mAP, respectively. Comparison results show that our C²DFF-Net model achieves competitive detection accuracy with the ICAFusion model. However, our model is significantly more lightweight, with a 94.5% reduction in parameter count and a 92.4% reduction in computational cost compared to ICAFusion.

2) Quantitative Evaluation: To provide a more intuitive comparison of detection performance, we select five different methods and our C²DFF-Net for qualitative comparison. Specifically, as shown in Fig. 9(a), the freight car target in the visible image at night is located in a dark area, and all methods fail to detect or misclassify the target, except for two infrared single-mode methods and our C²DFF-Net, which successfully detect the target. While infrared single-mode detection is unaffected by low-light conditions at night, the lack of texture information in infrared images often leads to confusion, especially for targets with similar contours. As shown in Fig. 9(b), both YOLOv8n and FFCA-YOLO suffer from misclassification in infrared detection mode. In addition, for the car target in Fig. 10(b), located in a region of strong light exposure, both FFCA-YOLO's visible single-mode detection and ICAFusion miss the target, while our C²DFF-Net successfully detects all targets. This is attributed to the incorporation of exposure and strong light effects in our ALM mask training strategy.

In addition, as shown by the truck target in the middle region of Fig. 10(a), due to the small size and color similarity to the background, all methods miss the detection except our C²DFF-Net. As seen in Fig. 10(b), the targets are so small and dense that even the human eye has difficulty distinguishing the target categories, and C²DFF-Net has only one missed detection, while the detection results of the other models are unsatisfactory. These visual comparative experiments demonstrate the robustness of C²DFF-Net to complex scenes and small targets that are dense, have strong light interference, or have similar colors to the background.

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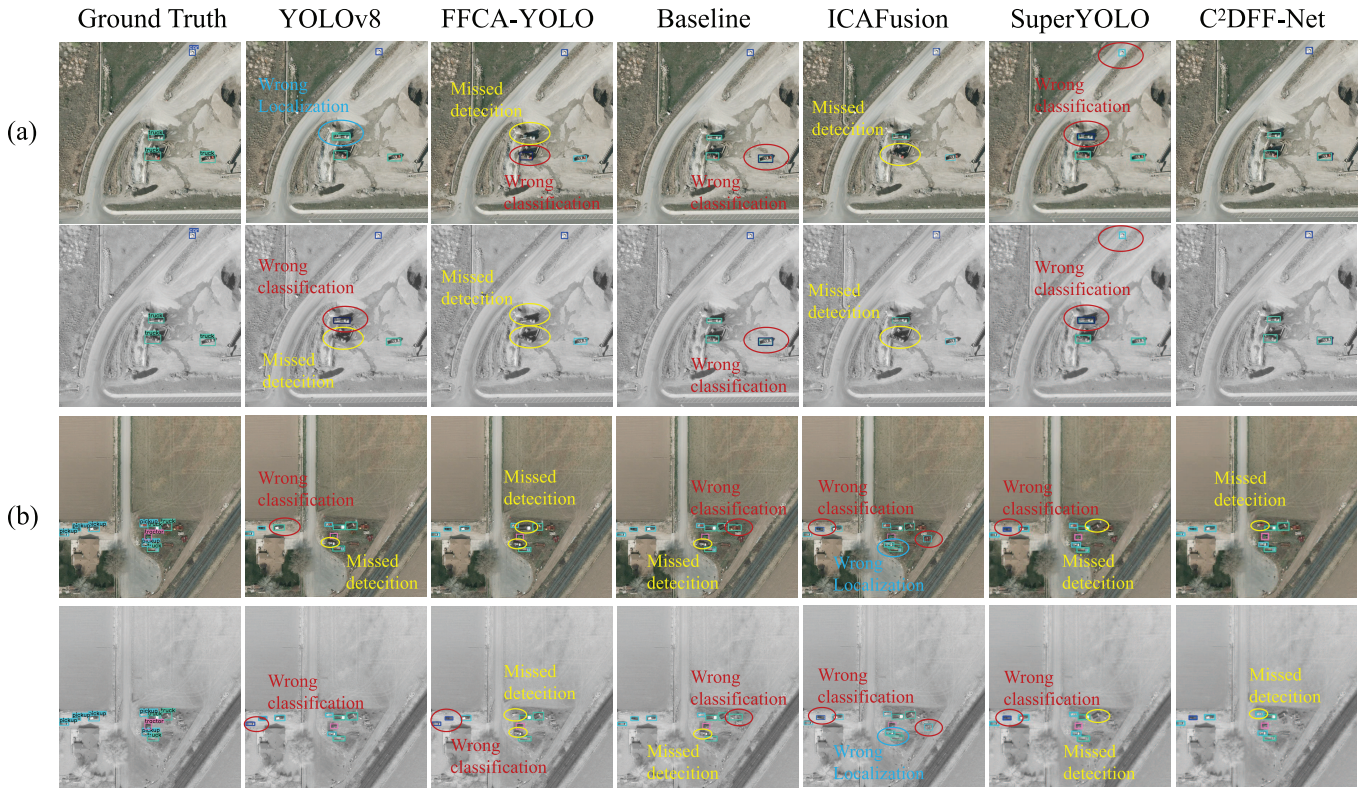


Fig. 10. Visualized detection results of different models on the VEDAI dataset. (a) Targets are small and have colors similar to the background, whereas in (b), they are small, dense, and difficult to distinguish.

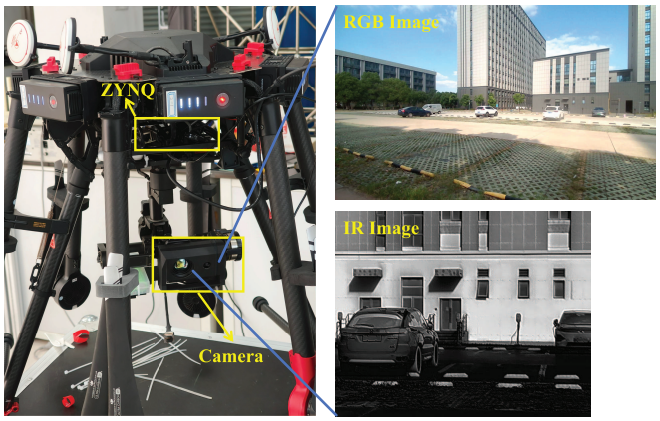


Fig. 11. Outfield test equipment and examples of captured images.

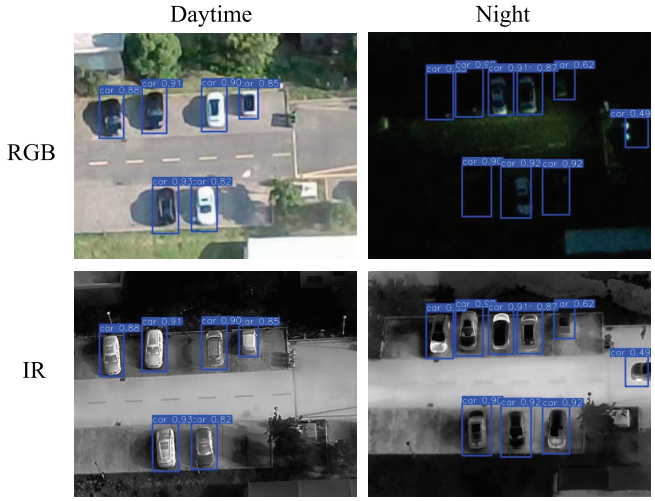


Fig. 12. Drone-based detection results for typical RGB-IR data pairs.

E. Real Scene Model Recognition Performance

To demonstrate the practical application capabilities of the proposed dual-modal model, we have conducted onboard target detection using a DJI M600 Pro drone equipped with a self-assembled dual-modal camera gimbal and a ZYNQ board. The experimental setup is shown in Fig. 11. The infrared camera is a Mini 640 from IRay, with a focal length of 25 mm and an image resolution of 640 × 512 pixels. The visible light camera is an industrial-grade HD camera with an image resolution of 1920 × 1080 pixels. Both cameras are connected to the ZYNQ via USB cables for synchronized capture and image registration, ensuring consistent resolution and pixel alignment. The registered image resolution is 640 × 512 pixels.

The drone-based detection results using our C^{2}DFF-Net model on high-altitude imagery are shown in Fig. 12. The results demonstrate that our method achieves satisfactory performance under various lighting conditions and supports real-time inference for mobile deployment. Notably, even in visible nighttime images where some vehicles blend into the background, the algorithm maintains remarkable robustness.

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V. Conclusion

In this article, we propose an extremely lightweight multimodal fusion detector for remote sensing small target recognition, named C²DFF-Net. CDFIM and space-frequency cross-domain DF fusion are achieved by incorporating CDFIM and CGSA modules into the network backbone, respectively. Furthermore, we design an ALM module during the training stage to guide the model in learning multimodal complementary features through asymmetric masks, addressing the challenges posed by poor illumination. Both CDFIM and CGSA are standalone plug-and-play modules that can be embedded into any dual-stream detector, thereby enhancing the representation of local and global complementary features with minimal computational overhead. The ALM module can also serve as a training strategy for other bimodal detection models to mitigate the network's bias toward a single modality. Thus, our framework and modules offer a generalized solution with significant potential for multimodal detection tasks.

Qualitative and quantitative experiments on two multispectral remote sensing target detection datasets, DroneVehicle and VEDAI, as well as a generalized multimodal dataset, FLIR, demonstrate that our method outperforms advanced single and multimodal target detectors. It enhances the detection accuracy and robustness of remote sensing small targets in complex background and illumination conditions while maintaining a very low computational overhead, facilitating fast inference and deployment onboard. Particularly on the DroneVehicle, it establishes a new benchmark for multispectral object detection. Although our method demonstrates strong potential in multimodal small object detection tasks, several limitations remain and offer room for further optimization.

1) Currently, the detection model can only efficiently process multimodal images that have undergone spatio-temporal registration; thus, the overall detection speed and accuracy are influenced by the performance of the image registration algorithm. In future work, we plan to explore an end-to-end multimodal registration-and-detection framework to address the impact of weak alignment on detection performance and to improve the overall efficiency of matching and detection.
2) While our UAV deployment experiments demonstrate the feasibility and robustness of the model, a gap remains between its inference speed on embedded hardware and that on GPUs. Further research on hardware acceleration is needed to optimize both speed and memory utilization.

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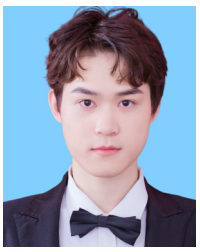
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